

Look-ahead Leakage in Open-Source LLM Trading Agents — Short Report

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*Backtesting an LLM trading agent on a window that lies **inside its pre-training corpus** is a silent look-ahead bias: the outcome is baked into the weights, and no chronological train/test split of the inputs can remove it. We call this **parametric look-ahead leakage**, and measure it with controlled, **price-only** trading agents over an in-distribution window (2024-H2) and a genuinely out-of-distribution one (2026).*

1. Problem and gap

Classical look-ahead bias is a *data* problem fixed by chronological splitting. An LLM adds a second, invisible channel — its **pre-training corpus**: if it covers the backtest window, the model can "recall" what happened and trade on it, and splitting the *input features* does nothing. Of the seminar papers, only **CausalStock** and **TradeMaster** mention look-ahead, both treating it purely as a splitting issue; **none** address a model whose pre-training already contains the test period. This study targets exactly that gap: *does an agent whose training data covers a window gain a non-causal edge there, versus (a) a model that cannot have seen it and (b) itself on a window it cannot have seen?*

2. Design

Three arms run an **identical Reflection + ReAct agent** on a **price-only causal context** (no news; a hard causality assertion) so any foresight must be *parametric*. Knowledge cutoffs are **measured by probe, not trusted from model cards**. The three arms (labels used throughout):

- **T-in** — *treatment, in-distribution*: qwen3:8b trading **2024-H2** (Jul–Dec), which it genuinely recalls — including **Aug-5** (the 2024-08-05 yen-carry-unwind crash, a sharp global selloff) and the June-2024 NVDA stock split.
- **C-A** — *model control*: llama3.1:8b (cutoff 2023-12) on the **same** 2024-H2 dates; it denies every 2024-H2 fact, so it cannot leak.
- **C-B** — *time control*: the **same** treatment model on **2026**, i.e. **out-of-distribution (OOD)** — strictly after its knowledge cutoff.

To strip out the 2024-vs-2026 **market-regime confound** we report a **difference-in-differences (DiD)** against an identical **no-memory momentum baseline**. Inference uses 3 seeds, a circular block bootstrap, permutation tests, and a **pseudo-event null** (98 random dates) for the crash-timing score.

3. Results

metric	T-in qwen3:8b (recalls)	C-A llama3.1:8b (control)	C-B qwen3:8b (OOD)
Sharpe	1.76	0.30	0.49
next-day prescience	+0.021	−0.032	+0.016
Aug-5 de-risk (pseudo-event p)	+0.115 (0.051)	−0.125 (0.92)	—

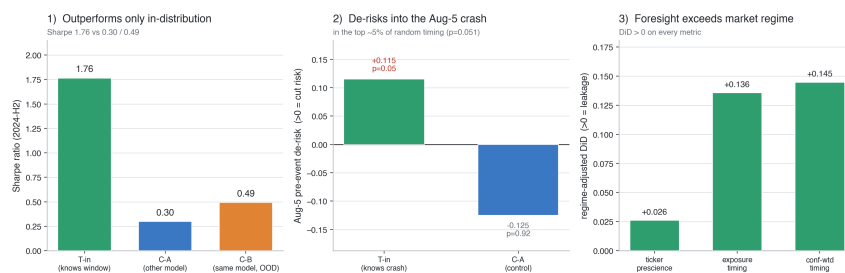
The knowledge-behaviour match is the core finding. The treatment cut risk into the Aug-5 crash it demonstrably remembers (+0.115; top ~5% of random-timing outcomes, $p=0.051$), while the control did the opposite (-0.125, $p=0.92$). It shows no election-timing edge — exactly the event the probe shows it does not surface — yet its book still significantly overweights the election winner JPM ($\Delta+0.088$, $p=0.001$): leakage that is behavioural even where verbal recall is suppressed. Behaviour mirrors *measured* memory item-by-item, which a generic "smarter model" or a regime artifact would not produce.

Regime is netted out, not the explanation. The regime-adjusted DiD is positive on every metric (exposure-timing +0.136); the no-memory baseline's own in-out gap is *negative*, so regime works *against* the result. Sharpe is descriptive — formal inference rests on the timing/DiD metrics, not raw Sharpe.

Family dependence (pre-registered). An independent-family co-treatment gemma3:12b (documented Aug-2024 cutoff) does not replicate the signature — it confabulates the period (projects "Biden" as the 2024 winner), with no crash de-risk ($p=0.54$). Lesson: a documented in-window cutoff is necessary but not sufficient — leakage needs *genuine recall*, so each model must be probed. An 8-model, two-size-tier sweep adds that, within every family, the larger model shows the larger regime-adjusted gap (qwen3 8B→32B +0.075→+0.133), though all per-model effects are individually non-significant.

Honesty about strength. Support is moderate and internally consistent but statistically modest at this sample size (cross-model $p\approx0.075$); the load-bearing evidence is the *pattern* — knowledge-matched timing, sign-consistent DiD, cross-model ordering — not any single p -value.

The leakage signature — a model that genuinely recalls the backtest window outperforms, de-risks before its known crash, and shows foresight beyond regime (only in-distribution)



4. Recommended robust-backtesting standards

(1) **Probe before you backtest** — empirically verify each model's knowledge of the test window; never trust documented cutoffs. (2) **Prefer out-of-distribution windows** strictly after the measured cutoff; treat in-distribution backtests as upper bounds. (3) **Use a model-control** with a verifiably earlier cutoff. (4) **Audit rationales**, not just returns, for confident confabulation. (5) **Report inference config** (thinking-mode, decoding, seeds) — it materially changes behaviour. (6) **Isolate the channel** (e.g. price-only) so the leakage source is unambiguous.